SALES ANALYSIS DASHBOARD

Project Documentation Report

Tool: Microsoft Power BI (Power Query + DAX)
Dataset: Sales_Analysis.xlsx, 9,994 order lines, 2016 to 2019
CoachX Data Analytics Project | October 2026

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1. Introduction

This project analyses the sales order records of an organization to give insight into sales performance and operations. The dataset holds order dates, customers, products, sales, profit, discount, shipping mode and regional data, which supports decisions on pricing, inventory, logistics and regional strategy.

Objectives

- Evaluate sales and profit over time to find trends.
- Identify top products, categories, customers, regions and cities.
- Assess shipping modes and their effect on sales and profit.
- Study the relationship between discount and profit to find a profitable discount strategy.

About the dataset

Item	Detail
Rows / columns	9,994 order lines / 21 columns
Period	January 2016 to December 2019
Customers / products / cities	793 / 1,850 / 531
Regions / segments	4 (Central, East, South, West) / 3 (Consumer, Corporate, Home Office)
Categories	Furniture, Office Supplies, Technology (17 sub-categories)

2. Methodology

The project follows an end-to-end Power BI workflow:

- **Load:** Sales sheet imported from Excel (Sheet2, 3 and 4 hold no analysis data and were not loaded).
- **Clean (Power Query):** converted dates to Date type, fixed Postal Code, kept Discount, Sales and Profit as decimals, removed Row ID.
- **Model:** created a Calendar table (Year, Month, Quarter, YearMonth), marked it as date table and related it to Order Date.
- **Measures (DAX):** Total Sales, Total Profit, Profit Margin %, Avg Discount, YoY Growth %, Running Total, Sales Contribution %, Top N ranking and others.
- **Visualize:** six report pages with consistent colours: Sales blue, Profit green, loss red.
- **Interactivity:** slicers (Year, Region, Segment), drill-down, drill-through, bookmarks and page navigation.

Data cleaning actions

Issue found	Action
Order Date and Ship Date stored as text	Converted to Date (English US locale)
11 blank Postal Codes (Burlington, VT)	Set type to Text and filled with 05401
Discount rounded to whole number (Avg 8.6%)	Restored Decimal type (correct Avg 15.6%)
Multiple lines per Order ID (5,009 unique)	Duplicate removal done only in a separate query

3. Requirement Analysis

The brief defines four tasks that the dashboard must satisfy:

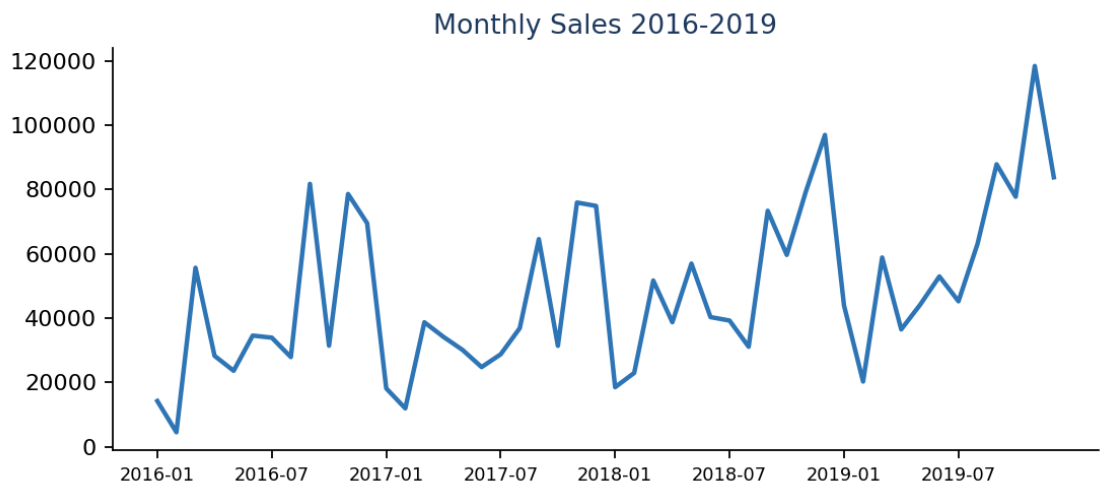
Task	Requirement	Delivered on
Task 1	10 charts: sales by customer, month, ship mode; profit by region; category; sub-category; dual axis; top 10 products; city map; discount scatter	Overview, Customer, Product, Time and Region, Discount pages
Task 2	10 calculations: totals, margin, discount, quantity, top 5 customers, YoY growth, ship mode sales	DAX measures and KPI cards
Task 3	10 charts: average and cumulative sales, sales by year and category, top 5 products by profit, segment and more	Customer, Product, Time and Region pages
Task 4	Power Query: 2019 filter, sales tax, remove duplicates, split name, group by year, sales category, discount filter	Power Query Analysis page

Other parameters

- KPIs: Total Sales 2.30M, Total Profit 286K, Profit Margin 12.5%, Quantity 37,873, Avg Discount 15.6%.
- Colour theme: navy #1F3A5F, blue #2E75B6, green #2BA58B, orange #F2A33A, red #D64550.
- Deliverables: Power BI file (.pbix), this report, and a presentation.

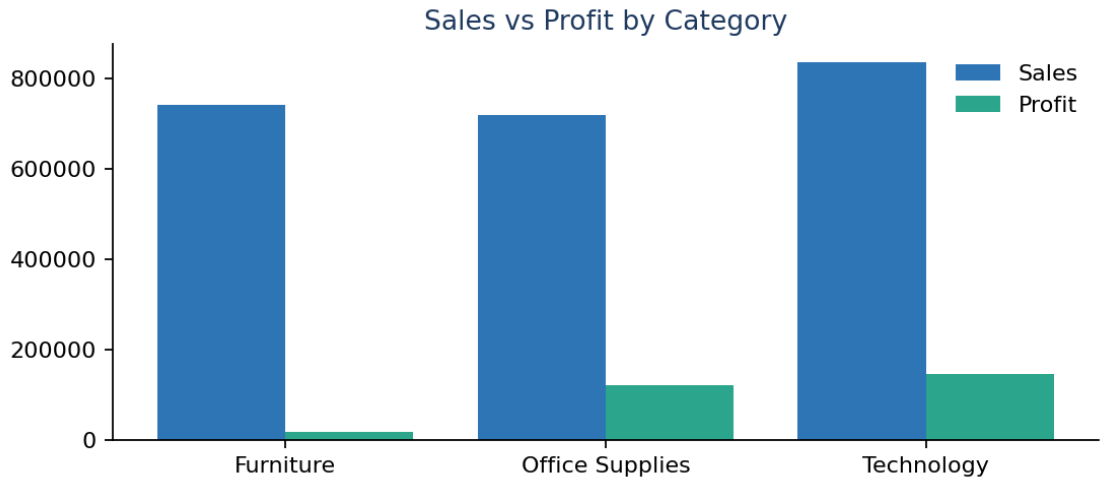
4. Visualizations and Insights

4.1 Sales trend



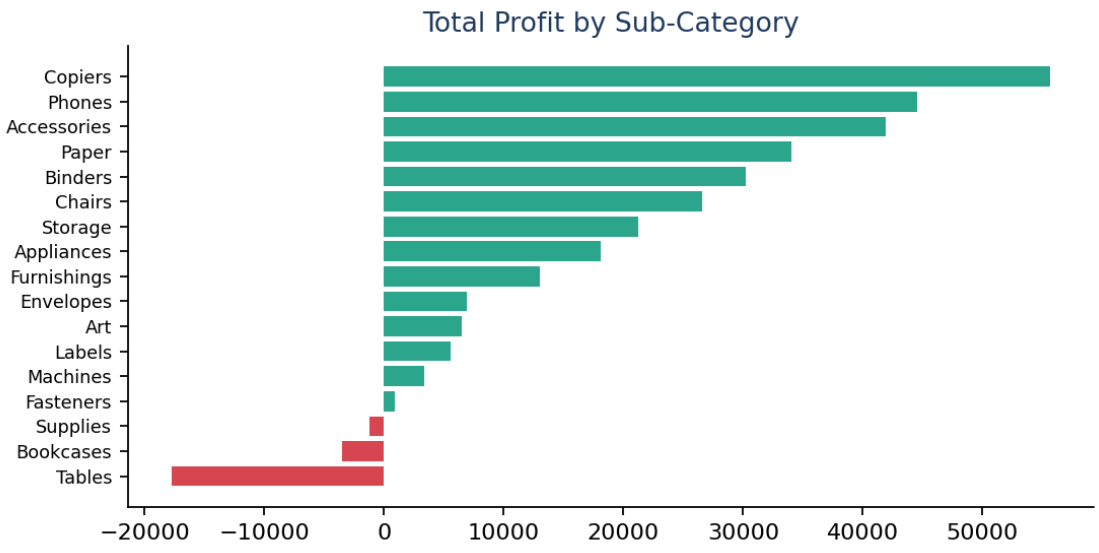
Total sales were 484K (2016), 471K (2017), 609K (2018) and 733K (2019). Year-over-year growth was -2.8% in 2017, +29.5% in 2018 and +20.4% in 2019. November and December are the strongest months.

4.2 Category performance



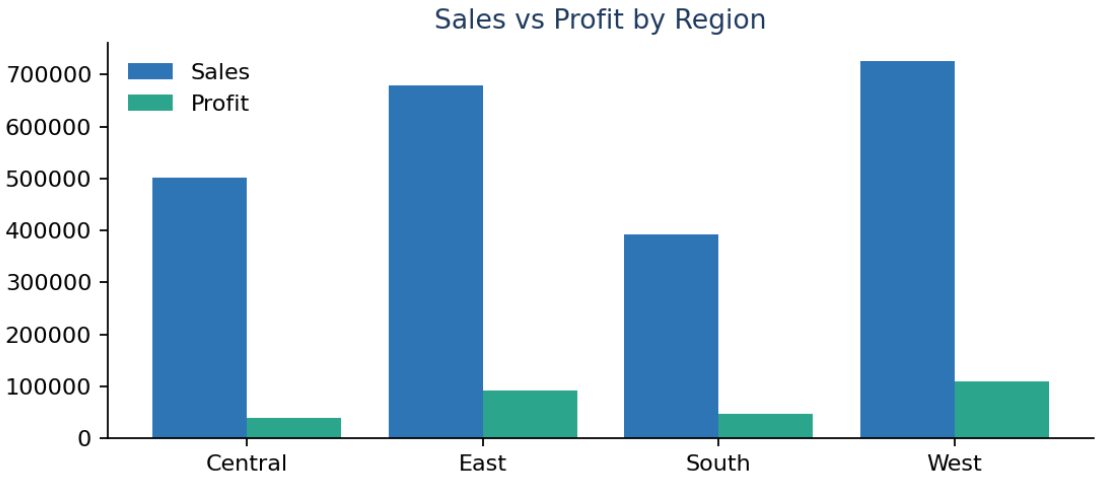
Technology earns the most (836K sales, 145K profit). Furniture has 742K sales but only 18K profit, a margin of about 2.5%.

4.3 Sub-category profit



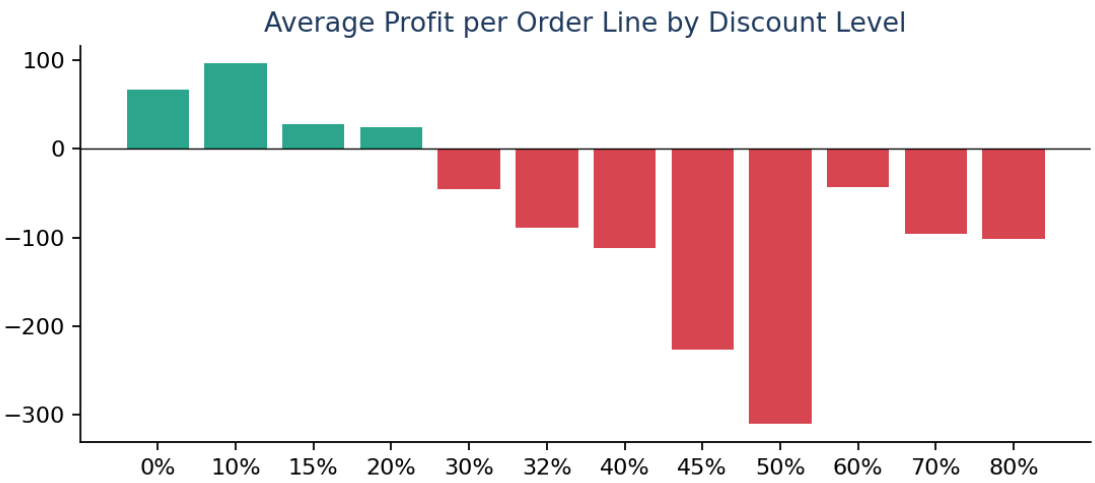
Tables (-17.7K), Bookcases and Supplies lose money, while Copiers and Phones earn the most.

4.4 Region



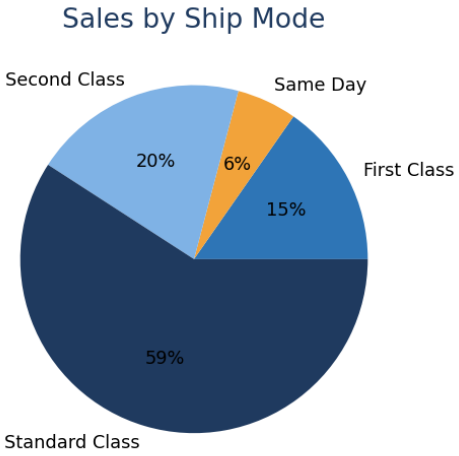
West leads with 725K sales and 108K profit. Central has 501K sales but only 40K profit, showing weak margins. West and East together give 61% of sales.

4.5 Discount and profit



Average profit per order line stays positive up to 20% discount and turns negative from 30%, reaching about -311 at 50%. Orders with a discount made a net loss of 34.6K, while orders without discount earned 321K.

4.6 Ship mode



Standard Class carries about 59% of sales, followed by Second Class (20%), First Class (15%) and Same Day (6%).

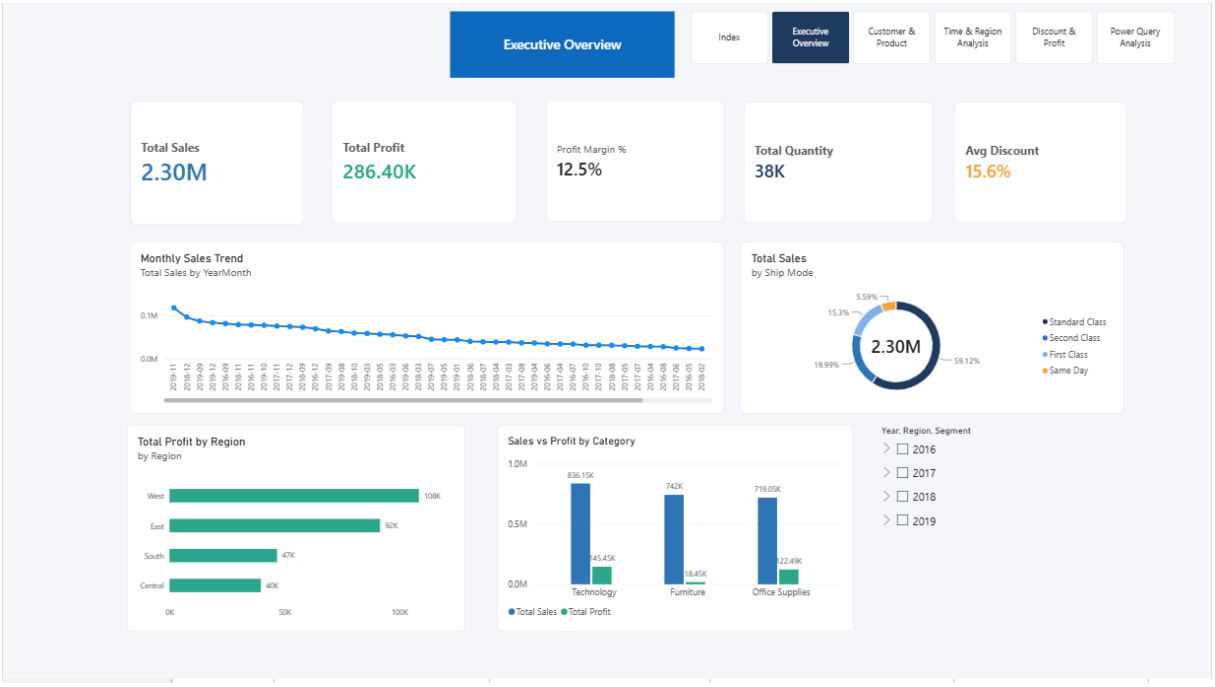
4.7 Customers, products and Power Query results

- Top customers: Sean Miller (25.0K), Tamara Chand (19.1K), Raymond Buch (15.1K).
- Top product: Canon imageCLASS 2200 Advanced Copier (61.6K sales, 25K profit).
- Segments: Consumer 1.16M, Corporate 706K, Home Office 430K.
- 2019 sales: 733K. Removing duplicate Order IDs cuts sales from 2.30M to 1.20M because one order holds several products.
- Average sales per order line fell from 243 (2016) to 221 (2019).
- Sales tax at 10%: Technology 83.6K, Furniture 74.2K, Office Supplies 71.9K.

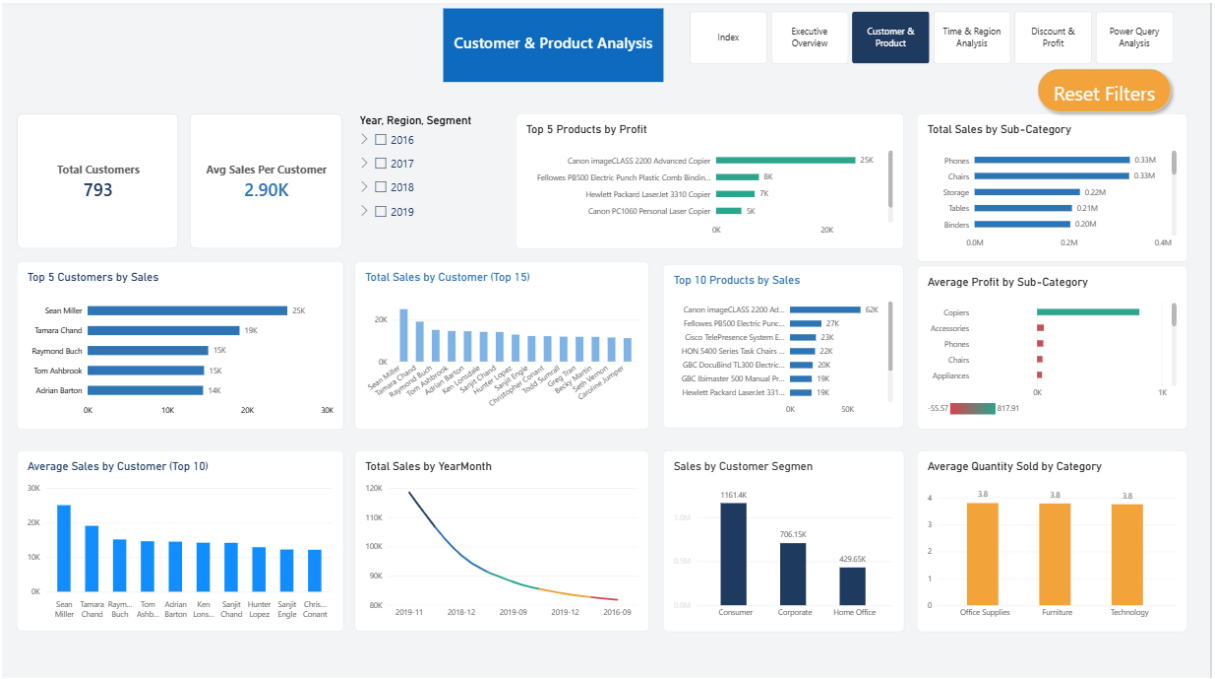
5. Dashboard Screenshots

The Power BI report has six pages (Index plus five analysis pages) with a navigation bar, slicers and a Reset Filters button.

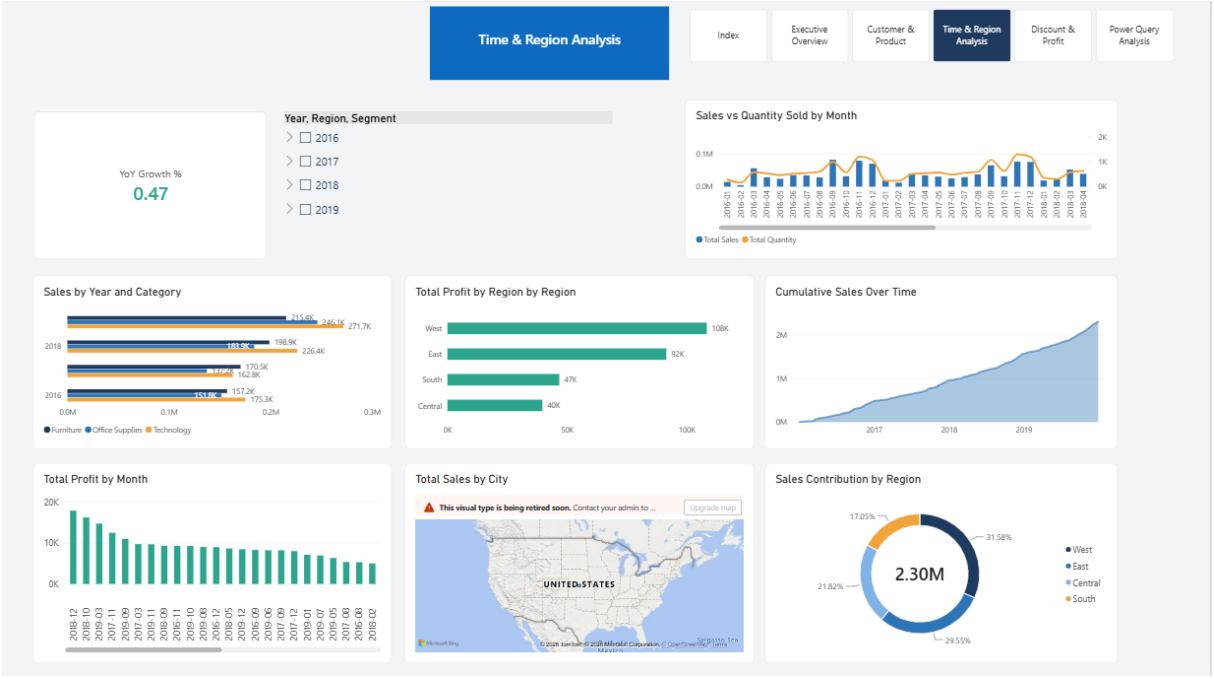
5.1 Executive Overview



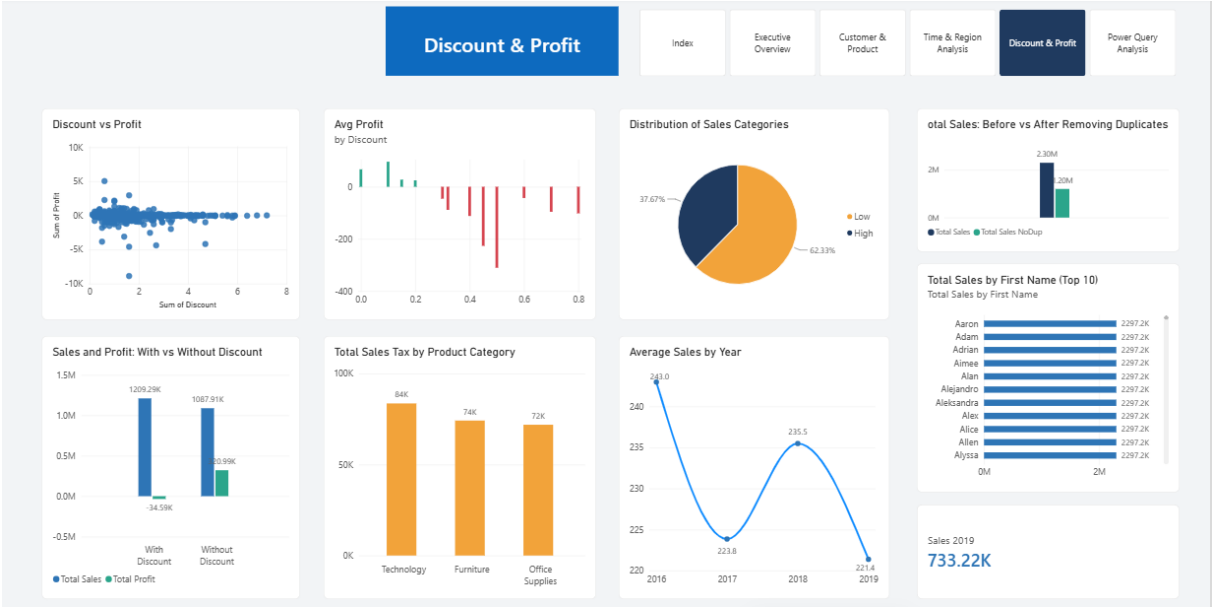
5.2 Customer and Product Analysis



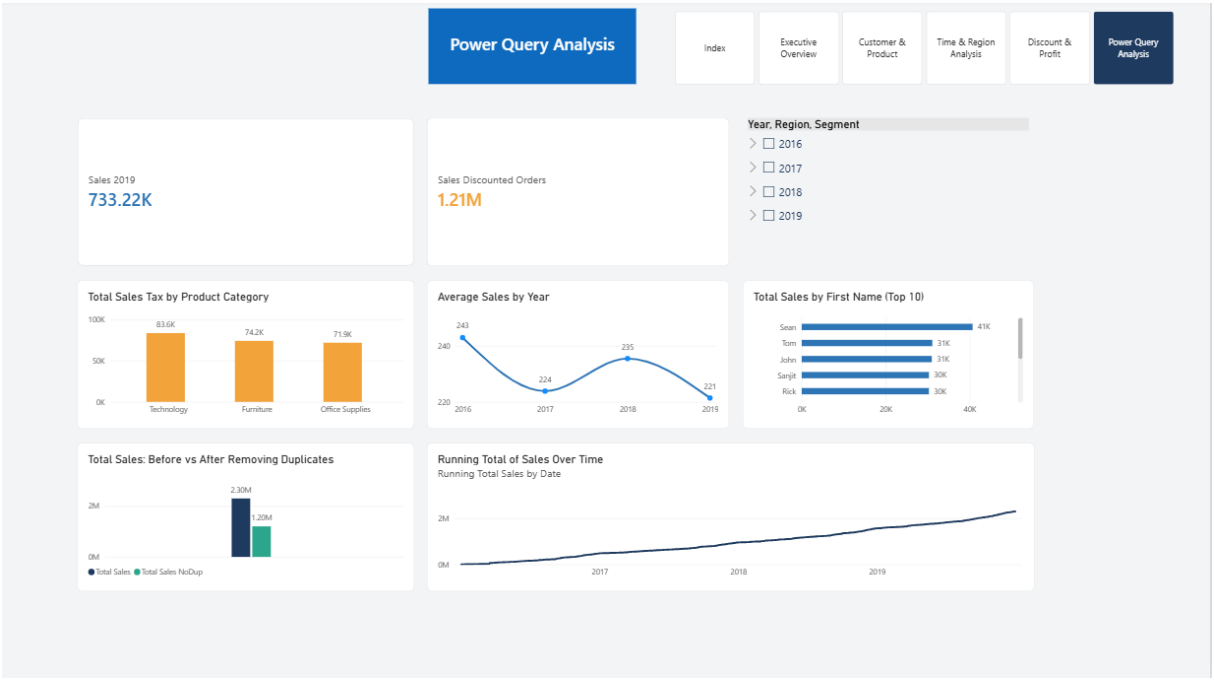
5.3 Time and Region Analysis



5.4 Discount and Profit



5.5 Power Query Analysis



6. Actionable Recommendations

- Cap discounts at 20 to 25%; losses start at 30%.
- Review pricing and sourcing for Tables, Bookcases and Supplies.
- Improve Furniture margin through pricing and cost control.
- Review Central region costs and discounting.
- Focus growth on Technology and the Consumer segment.
- Run a loyalty program for top customers.

7. Conclusion

The business grew steadily from 2016 to 2019 with a 12.5% overall margin. However, discounts above 20% convert profitable sales into losses, and Furniture plus a few sub-categories reduce profit. Controlling discounts and correcting pricing in weak areas can raise profitability without needing more sales volume. The interactive Power BI dashboard allows managers to explore these results by year, region and segment.